III Semester B.Tech., (CSE)
Data Analytics – CSS 2103
Internal Assessment4 - Mini Project Report
School of Computer Engineering
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On

"Data-Driven Study of AQI Variation in Indian Metropolitan Cities"

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1. Introduction

Air pollution has emerged as one of the most critical environmental and public health challenges of the 21st century. It directly affects human well-being, ecosystem balance, and urban liveability. Among the numerous indicators used to assess environmental quality, the **Air Quality Index (AQI)** serves as a comprehensive measure that reflects the combined impact of multiple pollutants such as **$PM_{2.5}$, PM_{10}, NO_2, SO_2, CO, O_3, and NH_3**. By translating complex air composition data into a single standardized metric, AQI provides an accessible summary of pollution severity and its potential health implications.

India's rapidly expanding urban centres—such as **Delhi, Mumbai, Chennai, Bengaluru, and Hyderabad**—have experienced increasing air quality concerns due to industrial growth, vehicular emissions, population density, and seasonal meteorological effects. Analysing AQI trends across these cities not only reveals the spatiotemporal variations in pollution levels but also reflects the broader environmental and socio-economic conditions shaping modern India's urban landscape.

Data analytics, therefore, plays a pivotal role in understanding and addressing these air quality issues. By leveraging historical pollutant data and computational models, analysts can uncover hidden patterns, identify critical pollution sources, and assess seasonal fluctuations with quantitative precision. This approach enables stakeholders—policy-makers, researchers, and citizens alike—to make informed decisions for sustainable urban management.

The motivation behind this study stems from the increasing urgency to apply **data-driven environmental intelligence** for tackling air pollution. In a country witnessing rapid urbanization and climate transitions, developing analytical frameworks that systematically evaluate AQI trends empowers both researchers and authorities to design **targeted mitigation strategies**, promote **public awareness**, and support **evidence-based policy interventions** for cleaner, healthier cities.

2. Objectives

1. To analyse AQI data across major Indian cities — Delhi, Mumbai, Chennai, Bengaluru, and Hyderabad.

2. To study variations and trends of key pollutants like $PM_{2.5}$, PM_{10}, NO_2, SO_2, CO, O_3, and NH_3.

3. To visualize air quality patterns using charts and graphs for clear comparison.

4. To identify seasonal and regional differences affecting pollution levels.

5. To summarize city-wise air quality performance through statistical analysis.

6. To develop a data-driven framework for future forecasting and policy insights.

3. Problem Statement

Air pollution has become a major concern in Indian cities due to rapid urbanization and industrial growth. Despite the availability of air quality data, it often remains unorganized and difficult to interpret. There is a need for a structured, data-driven analysis to compare AQI trends across major cities, identify pollution patterns, and understand seasonal variations. This project aims to bridge that gap by using data analytics to visualize, compare, and interpret air quality levels for better environmental insights.

4. Literature Review / Background Study

In the evolving environmental landscape, **India has emerged as one of the most polluted yet rapidly developing nations of the 21st century**. With rising industrialization, urban expansion, and vehicle emissions, Indian cities consistently rank among the world's most polluted urban centres. According to global air quality assessments, over 20 of the world's 30 most polluted cities are located in India *(Source: WHO, 2023)*.

Over the past decade, **air pollution has become a critical public health issue**, contributing to millions of premature deaths and respiratory illnesses every year. The **Central Pollution Control Board (CPCB)** and **National Air Quality Index (NAQI)** frameworks were established to monitor and standardize the measurement of pollutants such as $PM_{2.5}$, PM_{10}, NO_2, SO_2, CO, O_3, and NH_3 across India *(Source: CPCB, 2019)*.

Technological advancements in **data analytics and environmental modelling** have enabled researchers to better interpret pollution patterns and identify sources of emission. Studies indicate that **meteorological factors** like temperature, humidity, and wind speed significantly influence AQI variability, especially in northern and coastal regions *(Source: IIT Delhi, 2022)*. The integration of real-time monitoring, remote sensing, and computational tools has further enhanced urban air quality management efforts *(Source: Ministry of Environment, 2024)*.

At the centre of this transformation lies the **Air Quality Index (AQI)** — a unified measure that translates multiple pollutant concentrations into an easy-to-understand scale reflecting the health impact of air pollution. By analysing AQI data across multiple cities, researchers can identify temporal trends, spatial differences, and the overall direction of India's environmental progress.

4.1 Importance of AQI in Urban Analysis

The **Air Quality Index (AQI)** simplifies complex pollutant data into a single value, making it easier to communicate health risks and compare city-level air conditions. It serves as an essential indicator for urban planning, public awareness, and policy intervention *(Source: CPCB, 2019)*.

4.2 Impact of Seasonal and Geographic Factors

Air quality in India shows strong **seasonal variability**, with northern cities suffering from winter smog due to low temperatures and high emissions, while coastal cities like Mumbai and Chennai experience better dispersion due to sea breezes *(Source: WHO, 2023)*. Geographic conditions and local climate significantly influence these patterns.

4.3 Role of Data Analytics in Air Quality Studies

Data analytics enables accurate interpretation of large-scale air pollution data, uncovering hidden patterns, relationships, and correlations among pollutants. Visualization tools and statistical techniques help in identifying pollution peaks, seasonal shifts, and inter-city differences, providing a scientific basis for decision-making *(Source: Ministry of Environment, 2024)*.

5. Methodology

5.1 Data Collection

The dataset used in this project was compiled from verified air quality records for five major Indian cities — **Delhi, Mumbai, Chennai, Bengaluru, and Hyderabad**.
It includes daily pollutant levels for **$PM_{2.5}$, PM_{10}, NO_2, SO_2, CO, O_3, and NH_3**, along with the **Air Quality Index (AQI)**.
The data was cleaned, standardized, and aggregated to daily and monthly levels to ensure accurate comparison across all cities.

5.2 Tools and Technologies Used

The project was implemented using the **Python** programming language with the following tools and libraries:

- **Pandas, NumPy:** for data cleaning, manipulation, and statistical analysis.
- **Matplotlib, Seaborn:** for generating plots and visualizations.
- **Jupyter Notebook:** for interactive coding and experimentation.
- **SciPy, StatsModels:** For statistical testing and hypothesis analysis.
- **Scikit-learn (PCA, StandardScaler):** For dimensionality reduction and pattern identification.

6. Implementation / Experimentation

6.1 – Implementation :

Step 1 - Data Acquisition

- Loaded city-wise CSVs (Delhi, Mumbai, Chennai, Bengaluru, Hyderabad) into pandas Data Frames.
- Parsed timestamps and ensured a common date/time format for all records.

Step 2 - Data Preprocessing and Standardization

- Normalized column names for pollutants (**PM2.5, PM10, NO2, SO2, CO, O3, NH3**) and AQI.
- Converted non-numeric entries to numeric and handled missing values appropriately.
- Aggregated data to **daily** and **monthly** levels for cross-city comparison.

Step 3 - Exploratory Data Analysis (EDA)

- Plotted **time-series trends** of AQI and pollutants for each city.
- Generated **distribution plots and boxplots** to identify data spread and extreme pollution events.
- Constructed a **correlation heatmap** to analyze relationships among pollutants.

Step 4 - Feature Engineering and Scaling

- Computed monthly averages and rolling summaries to identify long-term patterns.
- Standardized features using **StandardScaler** for further multivariate analysis.

Step 5 - Principal Component Analysis (PCA)

- Applied **PCA** on pollutant datasets to detect dominant variance directions and key influencing factors.
- Used PCA to highlight similarities and differences in pollution behaviour across cities.

Step 6 - Visualization and Summary Generation

- Created and saved visual outputs such as:
    - **City AQI trend plots**, **Boxplots**, **Bar charts**, and **Correlation heatmaps**.
    - **PCA plots** and **summary tables** for city-level averages and pollutant intensities.

Step 7 - Statistical Analysis

- Performed **t-tests**, **chi-square tests**, and basic regression checks using **SciPy** and **StatsModels**.
- Compared air quality during different periods (e.g., monsoon vs winter) to validate observed trends.

Step 8 - Time-Series Modelling (Exploratory)

- Implemented **ARIMA** models for AQI forecasting in selected cities.
- Interpreted model behaviour cautiously due to data limitations and seasonality.

6.2 – Statistics used :

Descriptive Statistics : To summarize and describe key characteristics of the AQI and pollutant data (mean, median, range, standard deviation).
Formula:

$$\text{Mean} = \frac{\sum x_i}{n}, \text{Std. Dev.} = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$$

Calculated average AQI and pollutant concentrations for each city to identify general air quality levels and variation.

Correlation Analysis: To measure the degree of linear relationship between two pollutants or between AQI and individual pollutants.
Formula:

$$r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}}$$

Determined how strongly pollutants such as $PM_{2.5}$, PM_{10}, and NO_2 are correlated with overall AQI levels across cities.

Principal Component Analysis (PCA): To reduce data dimensionality while retaining maximum variance and identifying the main contributing pollutants.
Formula:

$$Z = XW$$

where X is the standardized dataset and W is the matrix of eigenvectors.

Applied PCA on pollutant columns to identify dominant pollution factors and visualize city-level air quality patterns.

One-Sample t-Test: To test whether the sample mean significantly differs from a hypothesized population mean.
Formula:

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

where $\bar{x}$ is the sample mean, μ is the hypothesized mean, s is the standard deviation, and n is the number of observations.

Tested whether the mean AQI for a given city significantly deviates from the national average AQI level.

Chi-Square Test: To determine whether two categorical variables are independent.
Formula:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

where O = observed frequency and E = expected frequency.

Used to analyse dependence between air quality categories (Good, Moderate, Poor) and city regions.

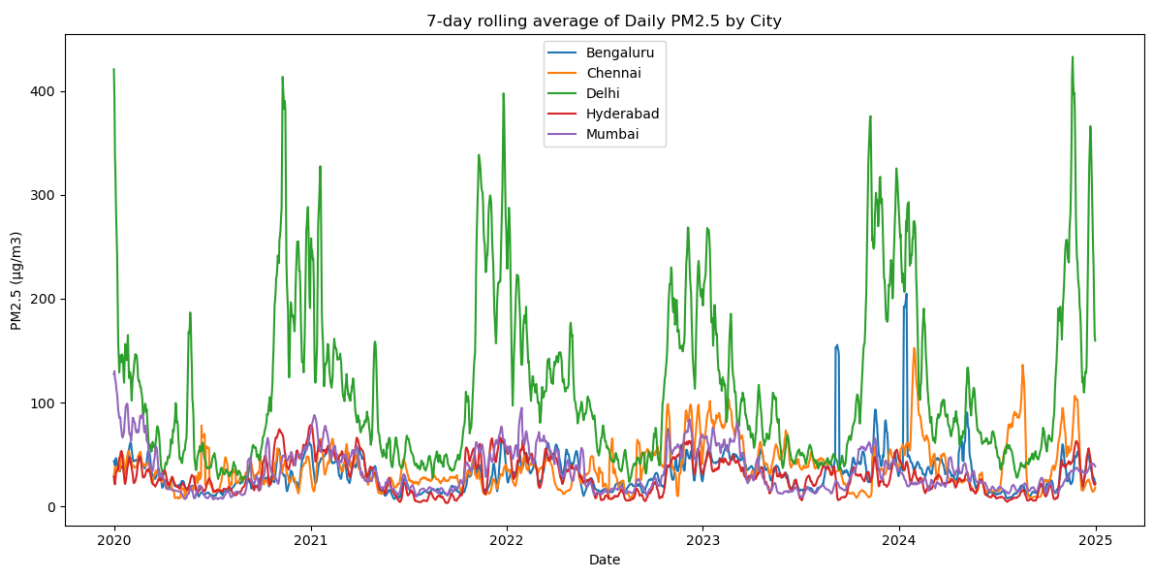
Time-Series Modelling (ARIMA): To analyse temporal dependencies and forecast future AQI trends based on historical data.
Formula:

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \cdots + \theta_1 \varepsilon_{t-1} + \varepsilon_t$$

Applied ARIMA models to predict short-term AQI variations and observe seasonal trends.

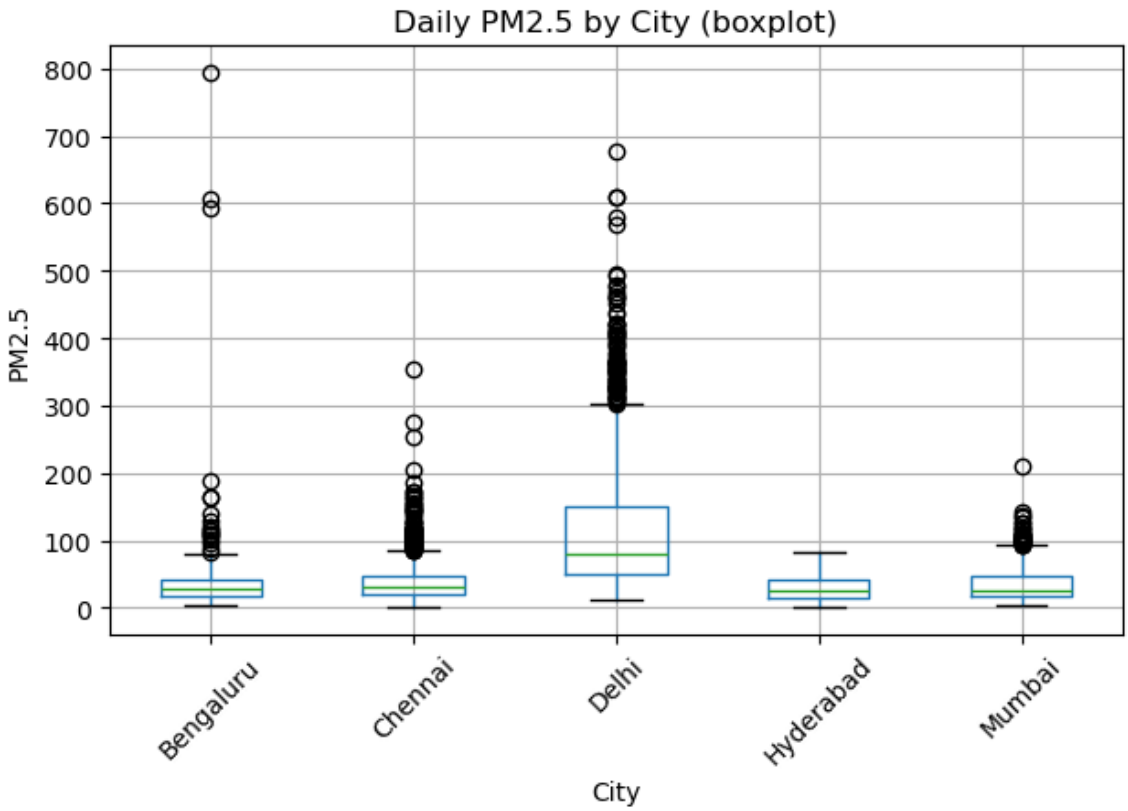
7. Results and Discussion

The analysis of AQI data across five major Indian cities — **Delhi, Mumbai, Chennai, Bengaluru, and Hyderabad** — revealed distinct variations in air quality patterns and pollution levels.



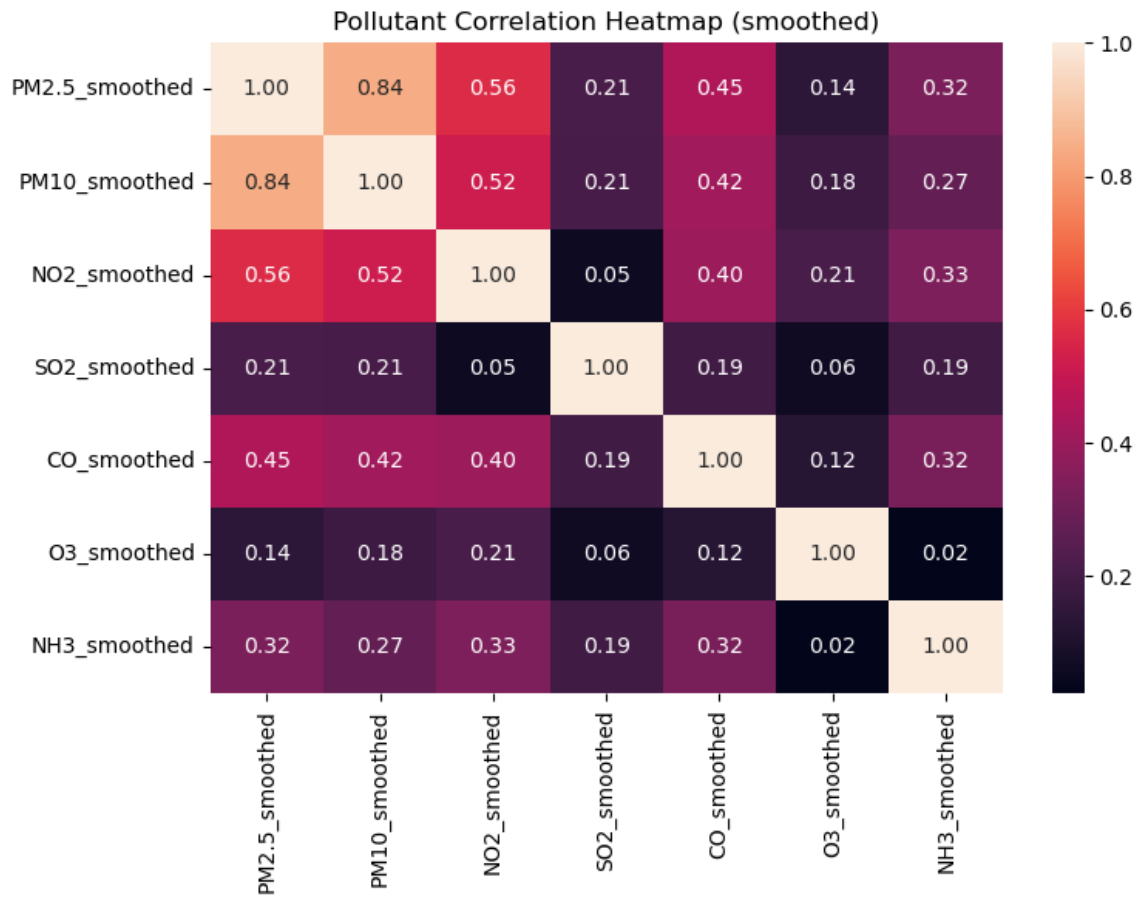
This line chart shows the 7-day rolling average of **$PM_{2.5}$ concentration** across five cities — Delhi, Mumbai, Chennai, Bengaluru, and Hyderabad — from 2020 to 2025.

- **Delhi** consistently records the highest $PM_{2.5}$ levels, especially during winter months.
- **Bengaluru**, **Chennai**, and **Hyderabad** show relatively lower values with smaller peaks.
- **Mumbai** exhibits moderate levels with occasional rises.
  → Indicates severe particulate pollution in Delhi and seasonal improvements in southern/coastal cities.



The boxplot compares the **distribution and spread** of daily $PM_{2.5}$ concentrations across cities.

- **Delhi** shows the widest box and many outliers, reflecting high pollution variability and frequent spikes.
- **Bengaluru**, **Chennai**, and **Hyderabad** show compact boxes, indicating stable and lower $PM_{2.5}$ levels.
- **Mumbai** lies in between, with moderate variation.
  → Confirms Delhi's extreme pollution compared to other metros and the relatively cleaner air in southern cities.



This heatmap visualizes the correlation among key air pollutants after smoothing the data.

- **$PM_{2.5}$ and PM_{10}** show a strong positive correlation (0.84), confirming that both vary similarly.
- **NO_2** and **CO** also show moderate correlation with PM values, suggesting shared emission sources (traffic/combustion).
- **O_3** and **SO_2** have weak correlations with others, indicating different formation mechanisms.

Two-Sample Z-Test (Delhi vs Mumbai AQI)

A two-sample **Z-Test** was conducted to compare the mean AQI levels of **Delhi** and **Mumbai**.
Outcome:

- Computed **Z-score = 35.53**, which is far greater than the critical value (±1.96).
- This indicates a **significant difference** between the mean AQI of Delhi and Mumbai.
  → **Delhi's air quality is statistically much worse** than Mumbai's.

Two-Sample T-Test (Welch's)

A **Welch's t-test** was used to validate the Z-test result without assuming equal variances.
Outcome:

- **t-statistic = 35.53**, **p-value ≈ 3.76×10^{-216}**, which is extremely small.
- The p-value < 0.05 confirms that the difference in mean AQI between Delhi and Mumbai is **highly significant**.
  → Confirms **Delhi's consistently higher pollution levels** compared to Mumbai.

Chi-Square Test (Pollution Category vs City)

A **Chi-Square Test of Independence** was conducted between **City** and **AQI Category** (Good, Moderate, Poor, Very Poor).

- **Chi^2 statistic = 2772.10**, **p-value = 0.0**, indicating strong dependence.
  → The distribution of AQI categories **varies significantly by city**, proving that air quality levels depend heavily on location.

Moving Average AQI Forecast (7 & 30 Days)

7-day and 30-day moving averages were computed to analyse short-term and long-term AQI trends for Delhi.

- The **7-day moving average** captures short-term fluctuations, while the **30-day average** smooths long-term variation.
- Both show a gradual **decline in AQI** over the final recorded period, suggesting mild improvement or seasonal clearing.
  → Indicates that **Delhi's pollution peaks are episodic**, with temporary recovery phases in between.

8. Conclusion and Future Work

The study successfully analysed AQI variation across five major Indian metropolitan cities—**Delhi, Mumbai, Chennai, Bengaluru, and Hyderabad**—using real air quality data and statistical methods.
Findings showed that **Delhi consistently exhibits the poorest air quality**, while **southern and coastal cities** maintain comparatively cleaner air.
Particulate matter (**$PM_{2.5}$ and PM_{10}**) was identified as the **dominant contributor** to AQI variation, with significant seasonal effects observed during winter.
Statistical tests confirmed strong **city-wise differences** in air quality, and visualization of pollutants provided clear evidence of inter-city variability.

Future Work:

- Extend analysis using **real-time AQI data** and integrate **meteorological parameters** such as humidity, wind speed, and temperature.
- Apply **machine learning models** for AQI forecasting and pollution source identification.
- Develop an **interactive dashboard** to monitor city-level AQI in real time.
- Conduct deeper **health-impact correlation studies** to link pollution exposure with public health indicators.

9. Team Member's Contribution

1. Member1 : Collected and cleaned city-wise AQI datasets (Delhi, Mumbai, Chennai, Bengaluru, Hyderabad) and standardized data formats for analysis.

2. Member2 : Performed data preprocessing, pollutant aggregation, and implemented statistical tests such as Z-Test, T-Test, and Chi-Square Test.

3. Member3 : Conducted Exploratory Data Analysis (EDA), generated trend graphs, boxplots, correlation heatmaps, and PCA visualizations

4. Member4 : Worked on time-series modeling using ARIMA and calculated moving averages for AQI forecasting (7-day and 30-day trends).

5. Member5 : Compiled results, interpreted outcomes, and prepared the structured IA4 project report including documentation and formatting.

10. References

1. **Central Pollution Control Board (CPCB)**, *National Air Quality Index (NAQI) – Technical Documentation*, Government of India, 2019.

2. **World Health Organization (WHO)**, *Global Air Quality Guidelines: Particulate Matter ($PM_{2.5}$ and PM_{10}), O_3, NO_2, SO_2, and CO*, 2021.

3. **Ministry of Environment, Forest and Climate Change (MoEFCC)**, *National Clean Air Programme (NCAP) Report*, 2022.

4. **Indian Institute of Technology Delhi (IIT Delhi)**, *Seasonal Variability of Air Pollution in Northern India: A Statistical Study*, 2022.

11. Appendix

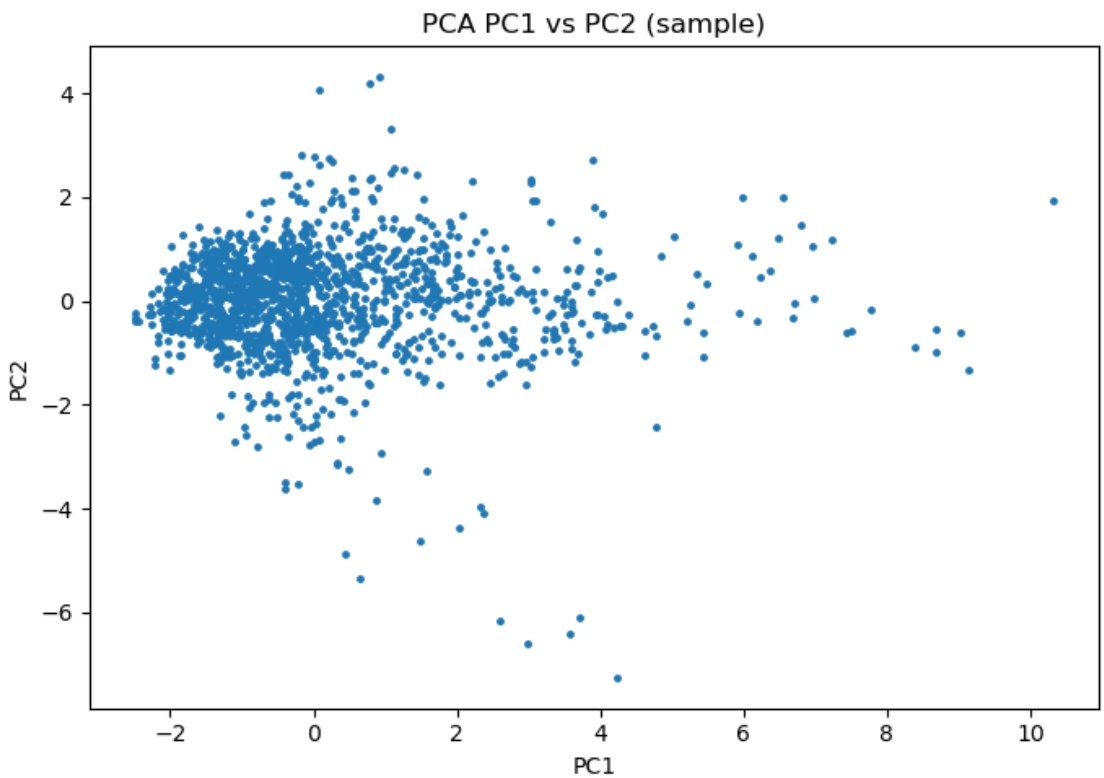
1. **Jupyter Notebook:** *DAGroup7CSEA.ipynb*

2. **Datasets Used:**

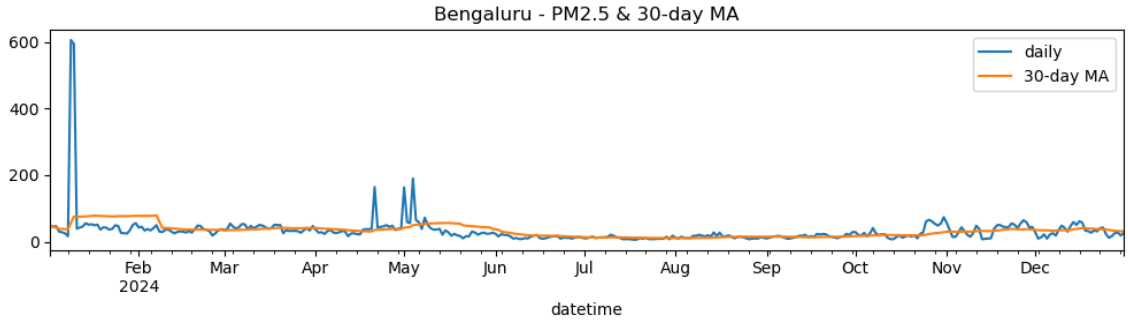
    - *delhi_combined.csv*
    - *mumbai_combined.csv*
    - *chennai_combined.csv*
    - *bengaluru_combined.csv*
    - *hyderabad_combined.csv*

3. Figures and Visualisations

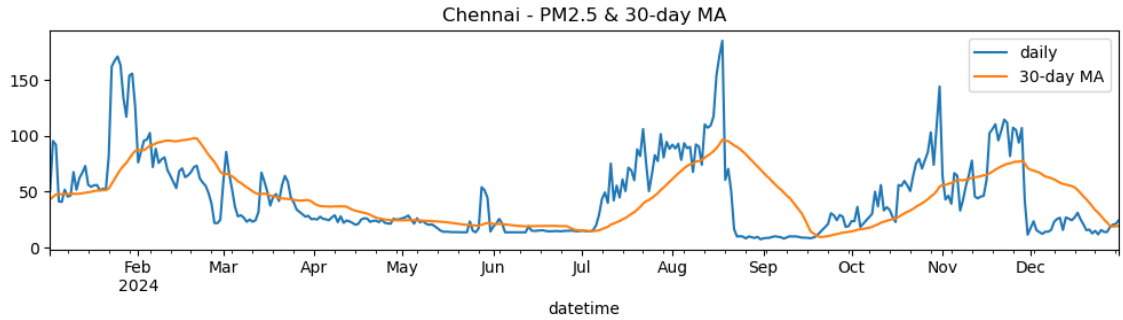
- Figure 1 - 7-day Rolling Average of Daily $PM_{2.5}$ by City
- Figure 2 - Daily $PM_{2.5}$ by City (Boxplot)
- Figure 3 - Pollutant Correlation Heatmap (Smoothed)
- Figure 4 - PCA – PC1 vs PC2 (Sample)
  The scatter plot shows the distribution of data along the first two principal components.
  It indicates that most variance is concentrated near the centre, showing limited outliers and confirming the dataset's compact structure.



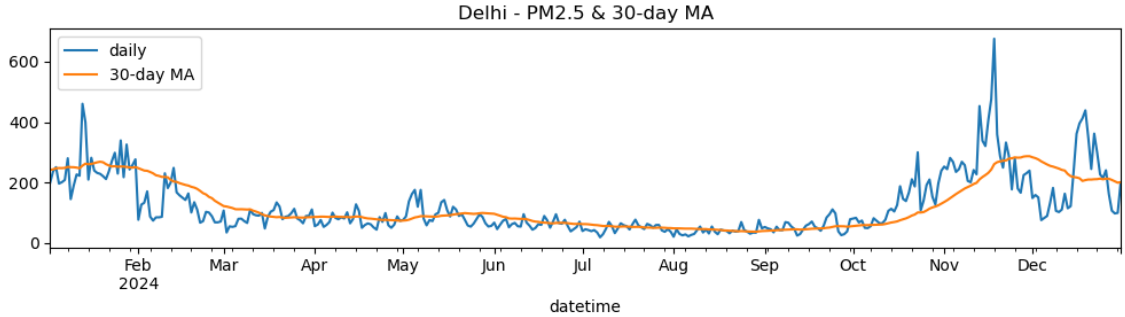
- Figure 5 - **Bengaluru – PM2.5 & 30-day Moving Average**
  $PM_{2.5}$ levels in Bengaluru stay low with occasional short spikes.
  The 30-day moving average shows overall stable and clean air quality.



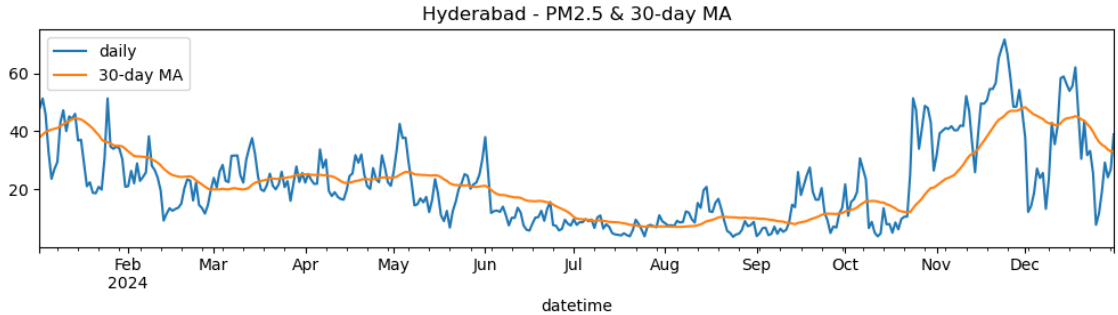
- Figure 6 - **Chennai – PM2.5 & 30-day Moving Average**
  Chennai's $PM_{2.5}$ shows visible seasonal peaks around August–December.
  The moving average confirms gradual pollution increase in late months.



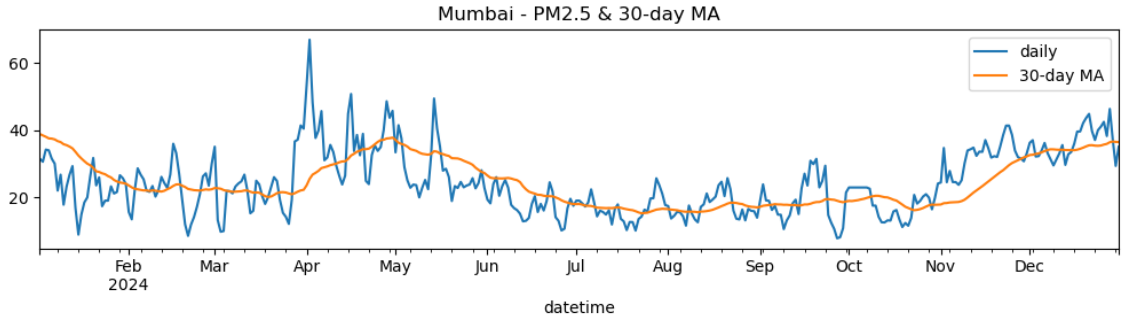
- Figure 7 - **Delhi – PM2.5 & 30-day Moving Average**
  Delhi records very high $PM_{2.5}$ levels, especially in winter.
  The 30-day average highlights severe and persistent pollution trends.



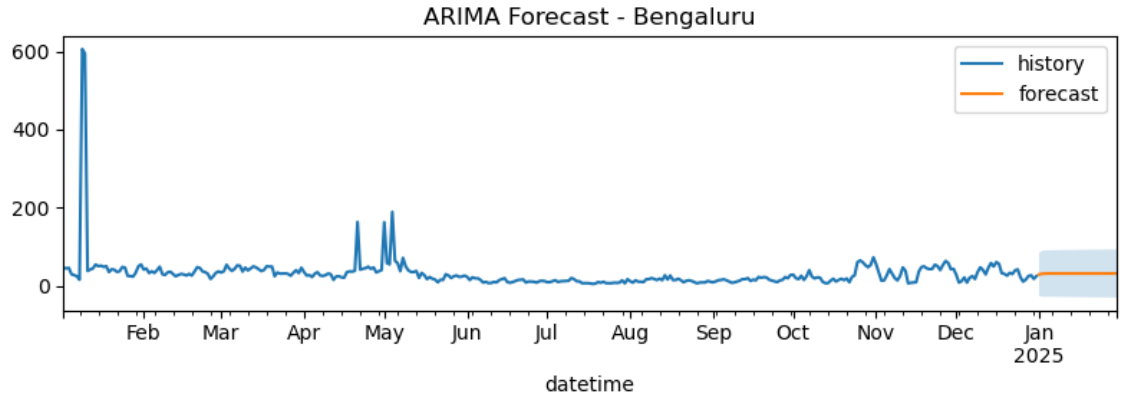
- Figure 8 - **Hyderabad – PM2.5 & 30-day Moving Average**
  Hyderabad experiences mild fluctuations in $PM_{2.5}$ with stable averages.
  Air quality dips slightly during winter, improving mid-year.



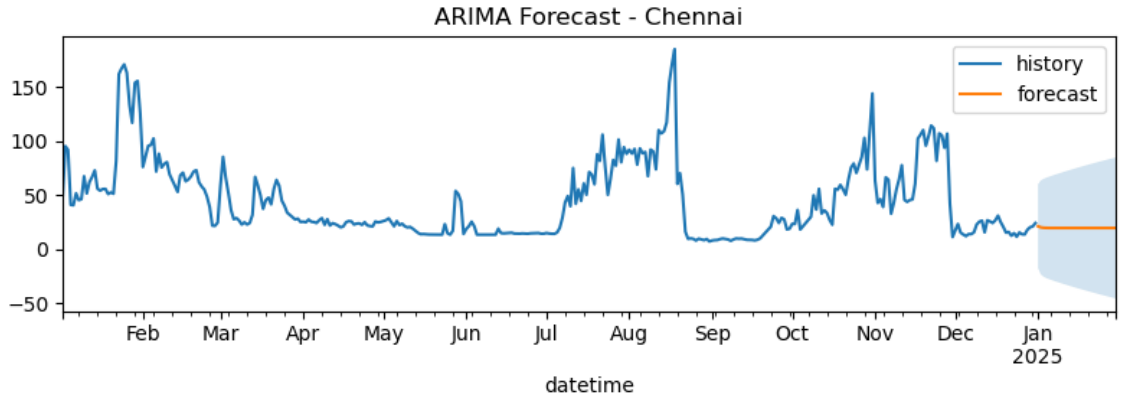
- Figure 9 - **Mumbai – PM2.5 & 30-day Moving Average**
  Mumbai maintains consistently low $PM_{2.5}$ levels year-round.
  The moving average indicates clean air with minimal seasonal change.



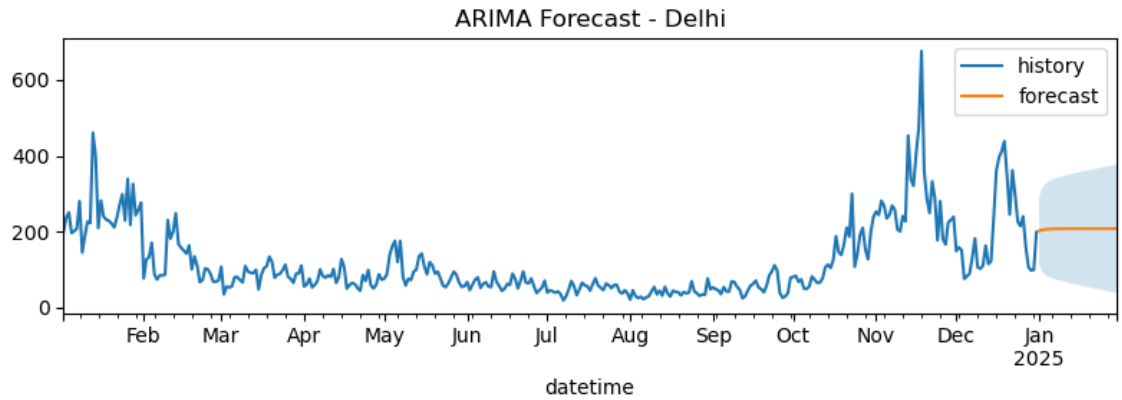
- Figure 10 - ARIMA Forecast – Bengaluru
  Forecast shows steady low $PM_{2.5}$ levels with minimal expected variation.



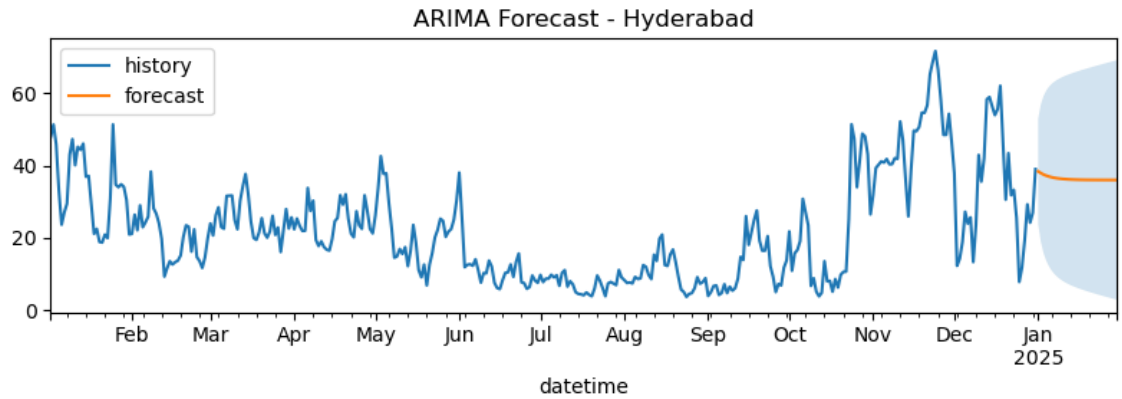
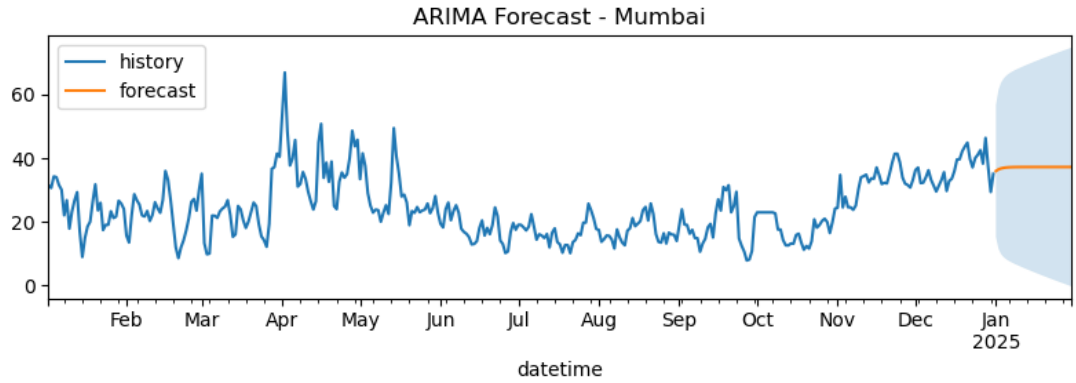
- Figure 11: ARIMA Forecast – Chennai
  Model predicts stable pollution trend with slight uncertainty bands.



- Figure 12: ARIMA Forecast – Delhi
  ARIMA forecast shows sustained high $PM_{2.5}$ levels and limited improvement.



- Figure 13: ARIMA Forecast – Hyderabad
  Forecast indicates moderate $PM_{2.5}$ levels with gradual downward trend.



- Figure 14: ARIMA Forecast – Mumbai
  Predicted $PM_{2.5}$ levels remain consistently low with minor fluctuations.